

apex industry association supporting electronics and IT hardware manufacturing industry, by 2020, overall market opportunity for supercapacitors in India will be approximately 1.3 billion units, with major demand from consumer electronics, EVs, renewable energy, railway, defence and more, accounting for the minimum revenue opportunity of ₹ 13.203 billion and the maximum revenue opportunity of ₹ 39.606 billion.

Market drivers

Supercapacitor technology has not yet been commercialised in the Indian market, and supercapacitors are not yet common in use. Only a handful of companies and research laboratories are working on their manufacturing. However, considering growing concerns of environmental issues, global warming, air pollution, green energy as well as government initiatives to make India a manufacturing hub, the Indian market is poised for an explosive demand for supercapacitors in various sectors due to their diverse applications in the following sectors.

Automotive. The supercapacitor market growth is most likely to be driven by automotive applications such as regenerative braking, start-and-stop systems and powering of on-board electronic features. Companies like Tata Motors and KPIT Technologies are already exploring supercapacitors as an alternate energy storage option for EVs.

Given the high growth projected for EVs under National Electric Mobility Mission Plan (NEMMP) 2020 in India, supercapacitors stand a strong chance to be utilised in this sector. NEMMP 2020 envisages the penetration of five to seven million EVs on Indian roads by 2020.

Indian Railways. Indian Railways is one of the world's largest rail networks, with metro trains also rapidly increasing in urban cities. With major thrust on safety, efficiency, energy conservation, power quality and freight wagon tracking, supercapacitors are poised to play a key role in Indian Railways.

Power recuperation and dynamic brake energy storage could be major applications. It is estimated that

TABLE I
OVERALL MARKET OPPORTUNITY FOR SUPERCAPACITORS IN INDIA BY 2020

Application	Total addressable market (million units)	Pessimistic forecast for market penetration (million units)	Optimistic forecast for market penetration (million units)
Automotive	720	72	216
Electric Vehicle	150	15	60
Railway	20	1.98	5.94
Wind Energy	0.56	0.11	0.17
Solar Energy	0.10	0.04	0.08
Solar Lighting	18	3.6	7.2
Consumer Electronics	372	18.6	37.2

Source: ELCINA

TABLE II
OVERALL REVENUE OPPORTUNITY (₹ BILLION) FOR SUPERCAPACITORS IN INDIA BY 2020

Application sectors	Revenue in pessimistic situation	Revenue in optimistic situation
Defence	0.00259	0.00518
Other application sectors	13.2	39.6
Total	13.203	39.606

Source: ELCINA

demand would reach around 10 million units of various supercapacitor modules in the next three years.

Renewable energy.

Wind energy sector. Across the world, wind turbines are, at present, using supercapacitors for wind turbine pitch control. In India, leading wind turbine manufacturer, Suzlon, is reportedly working with supercapacitors for their wind turbines. Increasing awareness and proven efficiency of supercapacitors for wind turbines across the world might trigger the usage of supercapacitors in India very soon.

It is estimated that the wind power potential of India is to the tune of 1GW to 5GW. In wind turbines, supercapacitor-based electrical pitch control can provide the best solution. Globally, more than 14,000 installed turbines are already using supercapacitors. It is anticipated that by 2020, supercapacitor penetration will increase by 30 per cent, for which an estimated 0.17 million units of supercapacitors will be required.

Solar energy sector. Solar PV panels and solar lighting are potential areas where supercapacitors can find immediate use in India. Considering India's

solar mission in place, the solar energy sector could act as an entry point for supercapacitors in the Indian market. With the launch of Jawaharlal Nehru National Solar Mission (JNNSM) in 2010, India is poised for the solar energy sector to take off.

In general, solar panels are used in both modes, online as well as offline, for which supercapacitors are strong options. With a conservative forecast of 40 per cent penetration of supercapacitors by 2020, demand in this sector is estimated to touch around 416,000 units. Various government-supported programmes like SHS, MNRE's solar lanterns and so on are estimated to create additional demand for supercapacitors, amounting to another 10 million units.

Consumer electronics. In India, since there are a limited number of consumer electronics manufacturers, opportunities for supercapacitors to be integrated directly into devices like cellphones and laptops is a bit difficult. A possible solution could be the power bank, wherein fast-charging and high-power-density features of supercapacitors can be well utilised.

Other opportunities in this sector